Project Report

This report presents the development and evaluation of three question-answering systems for Austrian tax law, covering EStG, KStG, UStG, and related legislation. Three approaches are compared: (1) direct inference with a large commercial language model, (2) fine-tuning an open-source model on curated domain data, and (3) retrieval-augmented generation (RAG) using official legal documents.

1. Models

Model 1 - Inference Only (GPT-4.1-mini)

The first system uses GPT-4.1-mini (OpenAI) accessed via the OpenAI API with no additional training or adaptation. GPT-4.1-mini is a compact, instruction-tuned variant of the GPT-4.1 family, supports up to 1 million tokens of context, and has a knowledge cutoff of June 2024.

Key hyperparameters and settings:

- The model was prompted in German to behave like an expert in Austrian tax law, answer concisely and formally, and cite legal provisions where possible.
- Temperature: 0 (greedy)
- max_completion_tokens: 200

The system prompt instructed the model to act as an expert in Austrian tax law ("Du bist ein Experte für österreichisches Steuerrecht"), to answer in formal German without preamble, to always cite relevant legal provisions in the format § [number] Abs. [x] [Gesetz], and to mention exceptions where applicable. Temperature was set to 0 to ensure reproducibility.

Model 2 - Fine-Tuned Model

The second system was based on ‘Qwen/Qwen2.5-1.5B-Instruct`. The Qwen model card describes it as an instruction-tuned causal language model with 1.54B parameters and a context length of 32,768 tokens.

Fine-tuning was performed in Google Colab using QLoRA with 4-bit quantization in order to fit the model into the free GPU environment. Instead of updating all model weights, a small LoRA adapter was trained, which made the setup more memory-efficient and feasible under Colab runtime limits.

The fine-tuning data combined two sources. First, a small public Austrian tax-law exam-preparation resource was used to collect concise legal question-answer pairs (20 QA pairs). Second, additional (approximately 72) synthetic training examples were created in the same style as the target task in order to expand coverage and encourage short German legal answers with paragraph citations. The intention was not to copy the evaluation set, but to create task-style examples that matched the format of the assignment more closely.

The lightweight QLoRA setup used the following hyperparameters:

- Base model: Qwen/Qwen2.5-1.5B-Instruct
- Quantization: 4-bit NF4 quantization
- LoRA rank: r=8
- LoRA alpha: 16
- LoRA target modules: q_proj, v_proj
- LoRA dropout: 0.05
- Training epochs: 1
- Batch size: 1
- Gradient accumulation: 16
- Learning rate: 2e-4
- Maximum sequence length: 256

Model 3 — Retrieval-Augmented Generation (RAG)

The third system implemented Retrieval-Augmented Generation. This approach augments the generation model with retrieved passages from external legal documents, so that answers are grounded in explicit legal context rather than only in the model's internal parametric knowledge.

The generator model was `Qwen/Qwen2.5-0.5B-Instruct`. According to the Qwen model card, this model has 0.49B parameters and a context length of 32,768 tokens. The retrieval model was `intfloat/multilingual-e5-base`, which the model card describes as a multilingual embedding model based on XLM-RoBERTa, with 12 layers, embedding size 768, support for 100 languages, and explicit training conventions that require "quer:" and "passage" prefixes for retrieval tasks.

The document corpus was assembled by downloading official PDF files of Austrian tax legislation directly from public government and legal information website RIS (Rechtsinformationssystem des Bundes). The following laws were included:

- Einkommensteuergesetz 1988 (EStG 1988)
- Körperschaftsteuergesetz 1988 (KStG 1988)
- Umsatzsteuergesetz 1994 (UStG 1994)
- Grunderwerbsteuergesetz 1987 (GrEStG 1987)

The PDF documents were converted to plain text and cleaned by removing repeated headers, footers, URLs, and print-preview artefacts. The text was then split into overlapping chunks. In the final RAG notebook, the chunk size was "800" characters with "100" characters overlap, the retriever returned the top "2" passages per question, and the generator was constrained to "80" new tokens.

2. Evaluation

The gold file (this is the test set with correct answers, that were made manually) contained expected answers and legal sources, which allowed evaluation not only of semantic similarity but also of legal citation correctness.

The main metrics are:

- BERTScore F1 - was used to measure how semantically close each generated answer was to the expected answer
- ROUGE-L - was included as a surface-level overlap metric that captures similarity in wording and answer structure

In addition, separate citation-based metrics were used, because a legal answer can sound plausible while still citing the wrong legal provision:

- Citation EM - checks whether the set of predicted legal citations exactly matches the set of gold citations (“1" is matched, ”0” is not matched, usually gives “0”, because of not 1:1 similarity)
- Citation Precision - which measures how many predicted citations were correct
- Citiation Recall - measures how many expected citations were recovered by the model
- Citation F1 - balances citation precision and recall

Two additional supplementary metrics were also included:

- Cosine Similarity - another semantic similarity measure based on text embeddings
- Readability - a descriptive stylistic measure of how easy the generated answers were to read

*Readability was not interpreted as a correctness metric, but only as an additional indicator of answer style.

3. Main results

Model	BERTScoreF1	ROUGE-L	Citation EM	Citation Precision	Citation Recall	Citation F1	Cosine Similarity	Readability
Inference	0.709	0.166	0.084	0.129	0.146	0.13	0.7	43.787
Fine-tuning	0.689	0.137	0.199	0.208	0.209	0.207	0.604	47.505
RAG	0.678	0.14	0.232	0.233	0.232	0.233	0.545	45.342

The best-performing model depends on the evaluation criterion.

If the main goal is overall answer quality and semantic similarity to the expected answer, the inference-only baseline performed best. It achieved the highest BERTScore F1, the highest ROUGE-L, and the highest cosine similarity. This indicates that its answers were overall closest to the expected answers in meaning and wording.

If the main goal is legal grounding and correct citation of legal provisions, the RAG model performed best. It achieved the highest Citation EM, Citation Precision, Citation Recall, and Citation F1. This suggests that retrieval from official legal documents improved the model's ability to refer to the correct legal basis.

The fine-tuned model ranked between the other two systems on most metrics. It improved citation behavior compared with the inference baseline, but it did not surpass the baseline on the main full-answer quality metrics. This suggests that lightweight fine-tuning with a small synthetic legal dataset improved domain adaptation somewhat, but was not strong enough to outperform the commercial inference baseline. However the Readability of this model is the highest overall.

4. Error Analysis

The three systems showed different types of errors.

The inference-only model generally produced the most fluent and semantically strong answers. However, it sometimes cited incorrect or incomplete legal provisions, and in some cases the wording was broader than the reference answer, which reduced citation accuracy even when the overall meaning was acceptable.

The fine-tuned model showed more legal adaptation than the inference baseline in citation-related behavior, but it was less stable in generation quality. The most common problems were incomplete answers, shorter but oversimplified responses, and occasional hallucinated or awkward legal phrasing. This suggests that the small synthetic training set and lightweight QLoRA setup were not sufficient to make the model consistently stronger than the baseline.

The RAG model performed best on legal grounding, but it was also the most sensitive to retrieval quality. When the retrieved passages were relevant, the answer often cited the correct law. However, when the retrieved context was noisy or only partially relevant, the final answer became too generic, incomplete, or grounded in the wrong legal section. This shows that RAG performance in this task depended strongly on preprocessing quality and retrieval precision.

Overall, the models did not make exactly the same mistakes. The inference model's main weakness was legal citation precision, the fine-tuned model's main weakness was unstable answer quality, and the RAG model's main weakness was dependency on retrieval quality and noisy PDF chunks.

For me, one of the main lessons of this project was that building legal QA systems is not only a question of model choice, but also of time, hardware, and experimentation strategy. I am still learninghow to build and evaluate such systems, and this project made me much more aware of the typical weaknesses of small fine-tuning setups and of RAG pipelines under limited compute. Fine-tuning and especially RAG preprocessing and generation took a long time, and Google Colab free-tier limits made experimentation slower and less flexible than I had expected. Because of these limits, I see this work as a strong first step rather than a final optimized solution.

For future work I would focus on three main improvements. First, I would further clean and structure the legal source texts before retrieval, because the RAG model was highly sensitive to noisy PDF chunks. Second, I would systematically tune chunk size, overlap, top-k retrieval, and generation limits on a larger validation subset, since the exploratory RAG experiment already showed that such changes can improve answer quality. Third, I would expand and validate the fine-tuning dataset more carefully in order to improve output stability and reduce incomplete or oversimplified answers.

5. Exploratory experiment

As a small follow-up experiment, I also tested an extended RAG configuration on the first 250 examples only. In this version, I increased the amount of retrieved legal context by using a larger chunking setup, for example increasing the chunk size from 800 to 1200 characters and also the maximal new tokens, because the last time because of the small number the answers were cut at random places. The purpose of this experiment was to see whether richer legal context could improve answer quality. The updated run achieved a BERTScore F1 of 0.699 and a ROUGE-L of 0.172 on the 250-example subset. Because this experiment was conducted only on a smaller subset, these results are not directly comparable to the main full-test results, but they suggest that RAG performance is sensitive to retrieval and chunking design. This also shows that there is still substantial room for improvement, and that with more systematic experimentation even stronger legal QA systems could be developed.

These follow-up experiments suggest that the current RAG pipeline is only an initial version. In future work, I would further optimize chunking, retrieval depth, prompt design, and generation limits, and I would also test more carefully cleaned legal source texts. The results indicate that even relatively small changes in retrieval settings can influence answer quality. Therefore, I see clear potential for building substantially stronger models with more time, more compute, and more systematic experimentation.

Model	BERTScoreF1	ROUGE-L	Citation EM	Citation Precision	Citation Recall	Citation F1	Cosine Similarity	Readability
RAG	0.678	0.14	0.232	0.233	0.232	0.233	0.545	45.342
RAG Updated	0.699	0.172	0.216	0.224	0.224	0.223	0.633	40.461

6. Reflecion

The aim of this project was to see how different QA approaches have different strengths in Austrian legal question answering.

The inference-only baseline was the strongest model in terms of overall semantic answer quality. It remained the best general-purpose system and produced the most coherent responses.

The fine-tuned open-source model demonstrated that domain adaptation with QLoRA is feasible even under limited resources such as Google Colab. However, in this setup the improvements were not large enough to outperform the inference baseline.

The RAG model provided the strongest legal grounding and the best citation-based performance. This suggests that retrieval from official legal sources is valuable for legal QA, but only when the retrieval stage and text preprocessing are sufficiently precise.

Overall, the results suggest that Austrian legal QA should not be evaluated by only one metric. General semantic answer quality and legal citation correctness capture different aspects of performance, and both are necessary for a balanced evaluation.

7. Resources

- https://openai.com/index/gpt-4-1/ "Introducing GPT-4.1 in the API | OpenAI"
- https://huggingface.co/Qwen/Qwen2.5-1.5B-Instruct "Qwen/Qwen2.5-1.5B-Instruct · Hugging Face"
- https://huggingface.co/Qwen/Qwen2.5-0.5B-Instruct "Qwen/Qwen2.5-0.5B-Instruct · Hugging Face"
- https://huggingface.co/intfloat/multilingual-e5-base "intfloat/multilingual-e5-base · Hugging Face"